BUSFIN 8200: Problem Set 2

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Question 1

(a)

We have complete markets iff we can find a unique matrix Ω s.t. $X\Omega = I$, where

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}, \qquad \Omega = \begin{bmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{bmatrix},$$

and each w_i is a portfolio weight vector using A, B and C. For example, $(10, 20, 30)'$ is asset A's payoff across the three states.

$X\Omega = I$ means

$$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad Xw_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad Xw_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Because X is 3×3, $X\Omega = I$ has a unique solution iff X is invertible, and

$$X \text{ invertible} \iff \det(X) \neq 0 \iff \operatorname{rank}(X) = 3.$$

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}.$$

Row reduction:

$$\begin{aligned} \text{row } 3 - \text{row } 2: &\quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{exchange rows 1 and 2}: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 2 - \text{row } 1: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 3 - 3 \cdot \text{row } 1 - \text{row } 2: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Thus $\operatorname{rank}(X) = 3$ and X is invertible. $\Omega = X^{-1}$ exists and is unique, with $X\Omega = XX^{-1} = I$. Thus, the market is complete.

(b)

Let X be the matrix of $N = 3$ non-redundant asset payoffs, let P be the price vector of the original assets, and let $P_c = (q_1, q_2, q_3)'$ be the price vector of the contingent claims. Use the LOP to solve for P_c as a function of X and P:

$$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}, \qquad X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}.$$

The LOP says that two assets with identical payoffs in every state of nature must have the same price. Therefore,

$$X'P_c = P \quad \Longrightarrow \quad \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Dividing everything by 10:

$$\begin{cases} q_1 + 2q_2 + 3q_3 = 1.6 \\ q_1 + q_2 + q_3 = 0.8 \\ q_2 + q_3 = 0.6 \end{cases} \quad \Longrightarrow \quad \begin{cases} q_1 = 0.2 \\ q_2 = 0.4 \\ q_3 = 0.2 \end{cases}$$

Thus, $P_c = (0.2, 0.4, 0.2)'$.

(c)

$$P_D = \text{Payoff}(D) \cdot P_c = \begin{pmatrix} 30 & 20 & 10 \end{pmatrix} \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 6 + 8 + 2 = 16.$$

(d)

$$P(X) = E[MX] = \sum_{s=1}^{S} \pi(s) M(s) X(s).$$

Since $q_s = \pi(s)M(s)$, we have $M(s) = q_s/\pi(s)$. With

$$P_c = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix} \quad \Longrightarrow \quad M(s) = \begin{pmatrix} 0.8 \\ 0.8 \\ 0.8 \end{pmatrix}.$$

By definition, no arbitrage means that

1. An asset with payoff ≥ 0 must have price ≥ 0.
2. An asset with payoff ≥ 0 and a chance for payoff > 0 must have price > 0.

We have $X \geq 0$, $\Pr(X > 0) > 0$, and $P(X) = E[MX] > 0$ (since $M > 0$). Thus, there is no arbitrage.

Moreover, $P(X) = E[MX] = 0.8\,E[X]$. The SDF stays constant across the three states, which implies it is risk-neutral.

(e)

$$\begin{aligned} P_D = E[MX_D] &= \sum_{s=1}^{S} \pi(s) M(s) X_D(s) \\ &= \tfrac{1}{4} \cdot 0.8 \cdot 30 + \tfrac{1}{2} \cdot 0.8 \cdot 20 + \tfrac{1}{4} \cdot 0.8 \cdot 10 \\ &= 6 + 8 + 2 = 16. \end{aligned}$$

This is the same as in Q1(c).

Question 2

(a)

I construct the unconditional linear SDF as

$$\hat{M}_t^* = \hat{a} - \hat{b}' x R_t,$$

where $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$ and $\hat{a} = 1 + \hat{b}'\hat{\mu}$, imposing the normalization $E[M^*] = 1$.

Table 1 reports the estimated SDF coefficients. The estimated SDF satisfies the unconditional pricing conditions, with $\widehat{E}[\hat{M}^*] = 1$ and $\widehat{E}[\hat{M}^* x R_j] \approx 0$ for all 10 portfolios.

Figure 1 shows no clear monotonic relationship between the estimated b_j coefficients and book-to-market deciles. Instead, the coefficients fluctuate considerably across portfolios, with both positive and negative values.

This pattern arises because the coefficients are determined by both the expected excess returns and the covariance structure of the portfolios, as reflected in $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$. Therefore, portfolios with higher average excess returns do not necessarily receive larger positive coefficients.

Table 1: Estimated BM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to BM decile j; estimation uses decimal monthly excess returns.

$\hat{a}$	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$	$\hat{b}_4$	$\hat{b}_5$	$\hat{b}_6$	$\hat{b}_7$	$\hat{b}_8$	$\hat{b}_9$	$\hat{b}_{10}$
1.0368	-1.5850	3.5503	0.7104	-3.1122	1.1837	2.9972	-5.2931	2.4722	2.8775	-1.1771

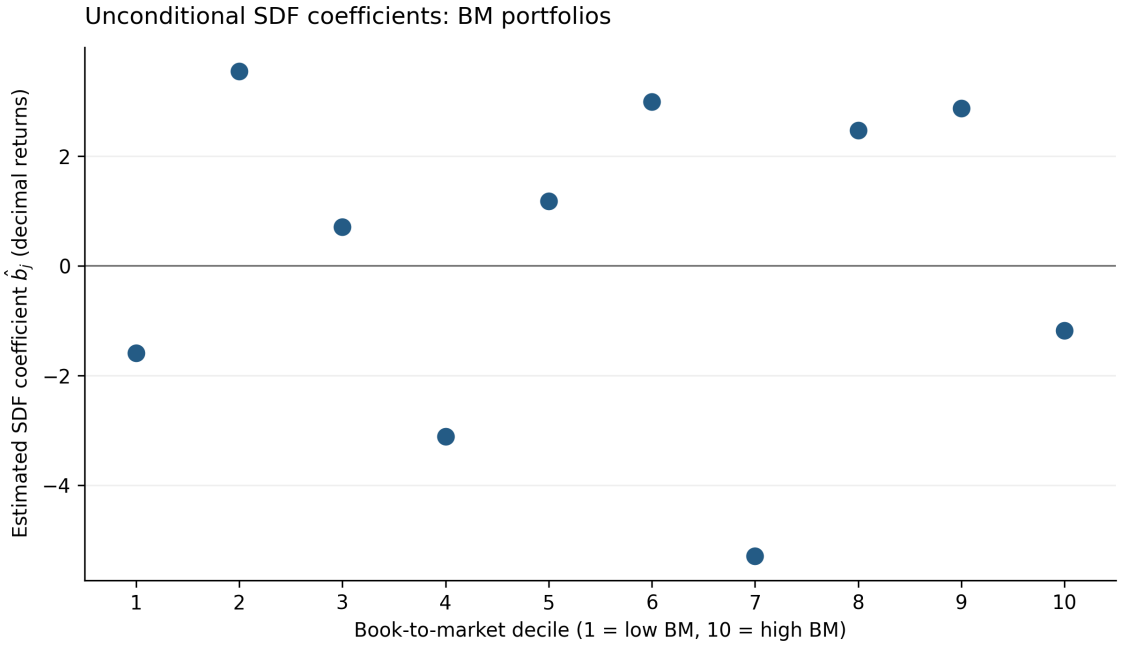


Figure 1: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor T.

(b)

Following the same approach as in part (a), I construct the unconditional linear SDF using the 10 momentum portfolios. Table 2 reports the estimated SDF coefficients.

Figure 2 shows that the estimated coefficients vary considerably across momentum deciles, with both positive and negative values. There is no clear monotonic pattern. As in part (a), this pattern reflects both the expected excess returns and the covariance structure of the portfolios.

The sample correlation between the BM and MOM SDFs is 0.3555, indicating a moderate positive relationship. This suggests that the two SDFs capture some common variation, but also reflect different components of asset returns. The sample correlation between the BM and MOM SDFs is

$$\widehat{\mathrm{Corr}}\left(\hat{M}^{*,BM}, \hat{M}^{*,MOM}\right) = 0.3555.$$

This uses 1,190 aligned monthly observations from July 1927 through August 2026.

Table 2: Estimated MOM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to MOM decile j; estimation uses decimal monthly excess returns.

$\hat{a}$	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$	$\hat{b}_4$	$\hat{b}_5$	$\hat{b}_6$	$\hat{b}_7$	$\hat{b}_8$	$\hat{b}_9$	$\hat{b}_{10}$
1.0666	-4.0194	1.1429	0.1024	1.4897	2.5317	-0.8195	0.2777	0.9818	-0.9414	3.4370

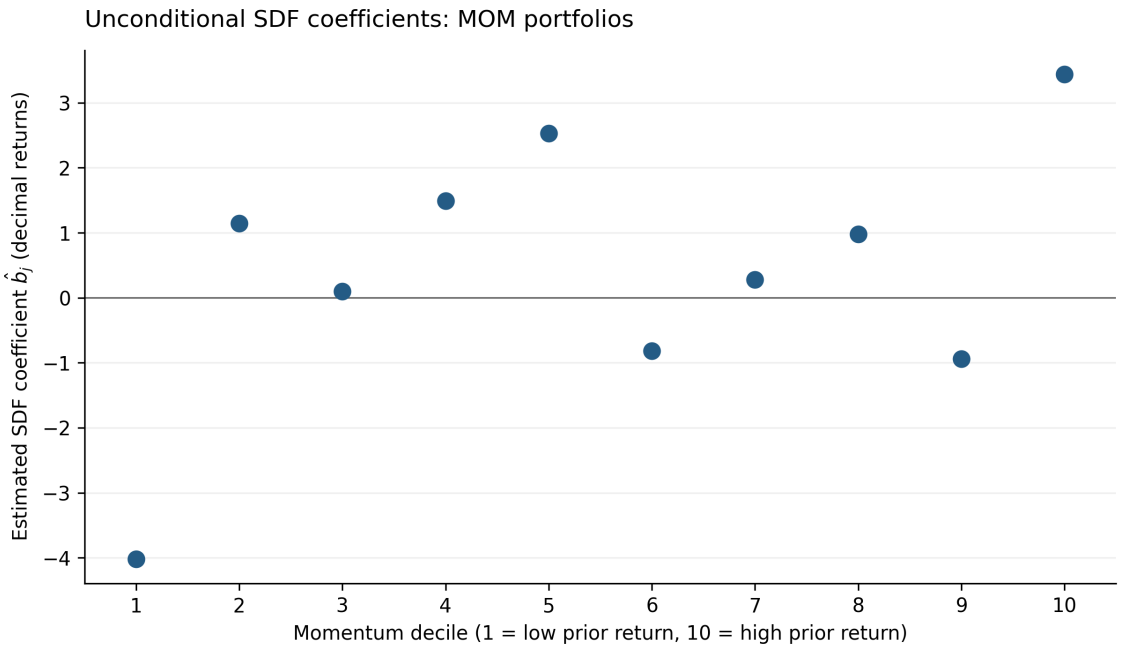


Figure 2: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$ for the ten momentum portfolios, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor T, as in part (a).

(c)

Following the same approach as in parts (a) and (b), I construct the unconditional linear SDF using the combined 20 BM and MOM portfolios. Table 3 reports the estimated SDF coefficients.

Figure 3 shows that the estimated coefficients vary considerably across both BM and MOM deciles, with positive and negative values and no clear monotonic pattern. Compared with parts (a) and (b), the coefficients change when the two sets of portfolios are combined, reflecting the additional covariance relationships and diversification opportunities.

Table 4 reports monthly Sharpe ratios of 0.1919, 0.2581, and 0.3045 for the BM, MOM, and combined maximum Sharpe ratio portfolios, respectively. The combined portfolio achieves the highest Sharpe ratio, consistent with the theoretical prediction that expanding the investment opportunity set cannot reduce the maximum attainable Sharpe ratio.

This result is also consistent with the SDF volatility bound discussed in class. Under the normalization $E[M^*] = 1$, the volatility of the constructed SDF equals the maximum Sharpe ratio. Thus, including both BM and MOM portfolios increases the maximum attainable Sharpe ratio and the volatility of the corresponding SDF.

Table 3: Combined SDF intercept and coefficients, rounded to four decimal places. The first ten coefficients correspond to BM portfolios and the next ten to MOM portfolios. Estimation uses decimal excess returns.

SDF intercept $\hat{a}$										1.0927
Decile j	1	2	3	4	5	6	7	8	9	10
BM	-1.9664	2.4739	0.5781	-3.7439	1.0655	2.6840	-4.6456	2.4707	3.7398	-0.2415
MOM	-4.4169	0.6354	-0.1712	1.9480	1.4236	-0.8781	-0.1639	0.9802	-1.2708	3.4473

Table 4: Monthly Sharpe ratios of $xR_t^* = \hat{b}' xR_t$, using the same 1,190 months from July 1927 through August 2026. Variances use divisor T.

Portfolio set	Monthly Sharpe ratio
BM	0.1919
MOM	0.2581
Combined	0.3045

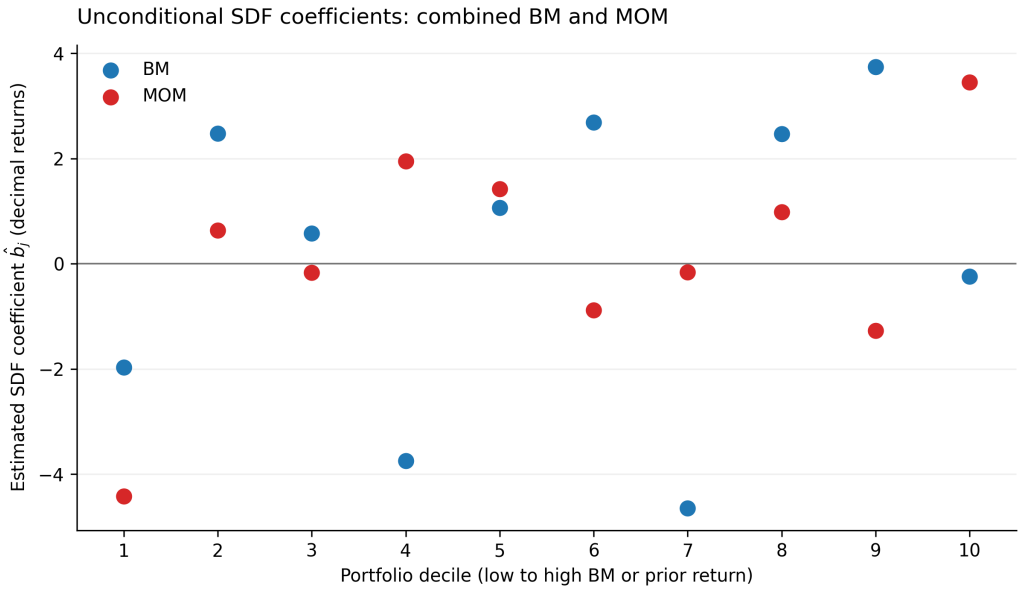


Figure 3: Coefficients in the combined BM–MOM SDF, normalized so that $\widehat{E}[\hat{M}^*] = 1$. Blue dots denote BM portfolios and red dots denote MOM portfolios. Returns are in decimals; the covariance matrix uses divisor T.

(d)

Table 5: Monthly pricing errors $\hat{\alpha}_p$ under the BM, MOM, and combined SDFs, in decimal return units and rounded to six decimal places. The common sample is July 1927 through August 2026 (1,190 months); covariances use divisor T.

Portfolio	BM SDF	MOM SDF	Combined SDF
BM: Lo 10	0.000000	-0.001511	0.000000
BM: 2-Dec	0.000000	-0.000101	0.000000
BM: 3-Dec	0.000000	-0.000223	0.000000
BM: 4-Dec	0.000000	-0.000778	0.000000
BM: 5-Dec	0.000000	0.000504	0.000000
BM: 6-Dec	0.000000	0.001261	0.000000
BM: 7-Dec	0.000000	0.000281	0.000000
BM: 8-Dec	0.000000	0.002376	0.000000
BM: 9-Dec	0.000000	0.003450	0.000000
BM: Hi 10	0.000000	0.003198	0.000000
MOM: Lo PRIOR	-0.010157	0.000000	0.000000
MOM: PRIOR 2	-0.005051	0.000000	0.000000
MOM: PRIOR 3	-0.003272	0.000000	0.000000
MOM: PRIOR 4	-0.001488	0.000000	0.000000
MOM: PRIOR 5	-0.000882	0.000000	0.000000
MOM: PRIOR 6	-0.000263	0.000000	0.000000
MOM: PRIOR 7	0.000509	0.000000	0.000000
MOM: PRIOR 8	0.001902	0.000000	0.000000
MOM: PRIOR 9	0.002240	0.000000	0.000000
MOM: Hi PRIOR	0.005418	0.000000	0.000000

The maximum absolute pricing errors are 8.67×10^{-17} for BM portfolios under the BM SDF, 3.64×10^{-17} for MOM portfolios under the MOM SDF, and 9.54×10^{-17} across all 20 portfolios under the combined SDF. All are below 10^{-12} in decimal return units.

For portfolios outside the group used to construct each SDF, the maximum absolute errors are 0.010157 for MOM portfolios under the BM SDF and 0.003450 for BM portfolios under the MOM SDF.